Spot Detection from Averaged TIFF Frames

Richardson–Lucy Deconvolution + Gaussian Component Fitting

Pipeline summary prepared for Overleaf from the uploaded Python workflow and result figures.

Objective and workflow

Goal

Detect point-like emitters in a noisy image stack, then decide whether each bright region contains **one** spot or **two overlapping** spots.

Processing pipeline

1. Load **50 cropped TIFF frames** and compute the average image.
2. Build a **Gaussian PSF** and apply Richardson–Lucy deconvolution.
3. Threshold the deconvolved image, label connected components, and crop local ROIs.
4. Fit **1-Gaussian** and **2-Gaussian** models; use BIC to keep the simpler or split model.

Key settings

Frames	50
RL iterations	10
PSF size	15×15
PSF σ	2.0 px
Threshold	0.02
ROI pad	4 px
BIC margin	25
Min separation	5 px

Main outcome

19 final spot centroids

Preprocessing and deconvolution

Stack averaging

$$\bar{I}(x,y) = \frac{1}{N}\sum_{k=1}^{N} I_k(x,y), \qquad N = 50$$

Averaging reduces frame-to-frame noise before spot enhancement.

Gaussian PSF + RL sharpening

$$\mathrm{PSF}(x,y) \propto \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right), \qquad \sigma = 2.0$$

The averaged image is normalized to $[0,1]$ and processed with **10 Richardson–Lucy iterations**.

Why this matters

Direct thresholding on raw averaged data can merge nearby emitters.

Deconvolution increases local contrast, making it easier to isolate candidate spots before model fitting.

Implementation note

RL is applied with a Gaussian PSF generated directly in the script.

Thresholding and connected-component detection

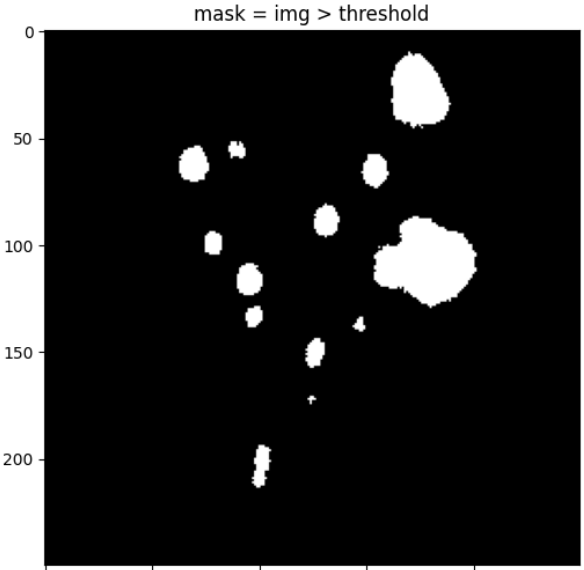
Binary mask

$$M(x, y) = \mathbf{1}[\mathrm{RL}(x, y) > 0.02]$$

Only pixels above threshold are retained as candidate signal.

Component filtering

- Connected components are labeled with `ndi.label`.
- Valid area range: **3 to 10,000 pixels**.
- Each accepted region is expanded by **4 pixels** on each side.



Local Gaussian fitting and model selection

Competing models per ROI

Single spot:

$$G_1 = A_1 e^{-\frac{(x-x_{01})^2+(y-y_{01})^2}{2s_1^2}} + B$$

Two spots:

$$G_2 = G_A + G_B + B$$

Each Gaussian has its own amplitude, center, and width.

Interpretation

A bright blob is split into two emitters only when the data support a meaningful improvement.

BIC-based decision rule

$$\mathrm{BIC} = n \ln\left(\frac{\mathrm{RSS}}{n}\right) + k \ln(n)$$

Choose the two-spot model only if

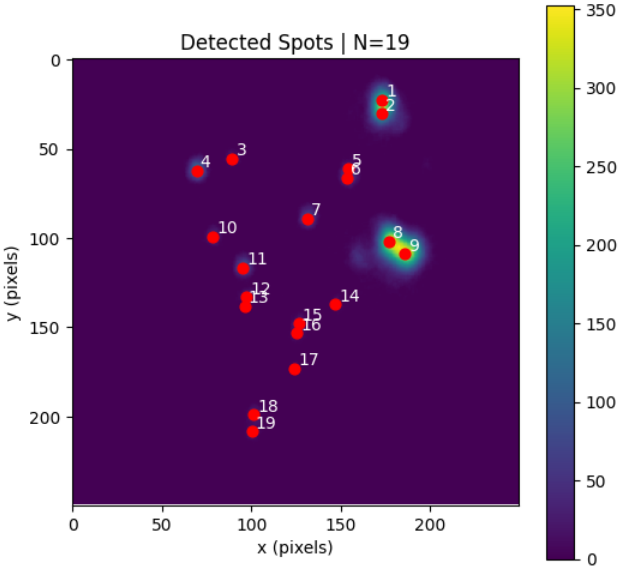
$$\mathrm{BIC}_2 < \mathrm{BIC}_1 - 25$$

and the fitted separation is at least **5 pixels**.

Why this helps

- Penalizes unnecessary extra parameters
- Reduces over-splitting while keeping overlaps resolvable

Detection results and next steps



Key observations

- Final count: **19 centroids**
- Larger bright regions can split into multiple emitters when the 2-Gaussian model is justified.
- The code checks close-pair distances for **(15,16)** and **(18,19)**.

Next steps

- Validate against manual labels or simulated ground truth.
- Test sensitivity to threshold, PSF width, and RL iterations.
- Export centroid tables for downstream analysis.